

Methods: PJ231356.6

ALMA Thesis Planner (automated pipeline)

0.1 1. Scope of the measurement

You are measuring two things, for every public target in ALMA project 2022.1.01311.S, and nothing else: (1) how the continuum flux density is distributed across angular scale, read directly off the calibrated visibilities; and (2) the peak rest-frame brightness temperature, read off the image-plane products. Both are morphology-agnostic and both are lens-model-free. You will not do parametric component fitting, source-plane reconstruction, or multiple-image identification, and every angular quantity you report stays in the observed frame. When you write the paper, say explicitly that converting observed-frame angular scales to source-plane physical scales requires a lens model, and that differential magnification is precisely why a compact fraction is an observed-frame quantity rather than an intrinsic one — that sentence belongs in the methods, not in a caveats appendix.

Two things about the proposal text you must not inherit uncritically. The abstract quotes "870micron/1mm" without labelling frame; these are ALMA receiver-band designations, so treat them as observed-frame and verify them against the archive's spectral-window metadata before you compute anything (Section 2). The abstract's 0.1" resolution and 100–500 pc scales are *requested/expected* values from the proposal, not delivered measurements; you will inventory the delivered beams yourself and quote those. Similarly, $z = 1-3$ is a stated sample range, not a per-source redshift, and $>10^{13} L_{\text{sun}}$ is described as an already-magnification-corrected intrinsic luminosity — use those exactly as the proposal frames them.

0.2 2. Weeks 1–3: archive retrieval, inventory, and pre-registration of the analysis choices

2.1 Retrieval. Query the ALMA Science Archive for project code 2022.1.01311.S using 'astroquery.alma' ('Alma.query(payload={'project_code': '2022.1.01311.S'})'), and pull the full list of Member OUSs, science targets, receiver bands, and the 'obs_release_date' field. The data description does not state proprietary status, so your first deliverable is a definitive, dated list of which targets are actually public. Whatever that number is, the census covers it; do not build the analysis around an assumed sample size. Download the raw ASDMs plus the pipeline 'scriptForPI.py' and restore the calibrated measurement sets in the CASA version named in the per-MOUS README (version matters — do not restore with a newer CASA), or request the calibrated MSs directly from the archive if delivery is offered. Also retrieve the pipeline weblog and the QA2 report for each MOUS; you will need the delivered beam, the achieved RMS, and the quoted absolute flux-scale accuracy from them.

2.2 Per-target inventory. For each MOUS run 'listobs' and record, in a machine-readable table (one YAML or CSV row per target, generated by script, never by hand):

- Receiver band and the observed sky frequency of each science spectral window (centre and

bandwidth), taken from `'msmetadata.chanfreqs()'`. This is your verification step for "870 μm / 1 mm": confirm that Band 7 windows sit near ~ 345 GHz and Band 6 windows near ~ 230 – 240 GHz observed, and quote the actual measured centre frequency per target in the paper, not the round number from the abstract.

- Number of execution blocks, on-source time, number of antennas, and the minimum and maximum projected baseline. Convert baselines from metres to $k\lambda$ per spw using the actual spw frequency (`'u $\lambda$  = u $_m$   $\nu/c$ '`) — never assume a single conversion for the whole project, since Band 6 and Band 7 targets will map the same physical array onto different uv distances.
- Delivered synthesized beam (major, minor, PA) and RMS from the pipeline continuum image.
- Redshift where the project metadata (archive source table, ancillary Planck/Herschel identifications carried in the project metadata) supplies one. Where it does not, the target carries the $z = 1$ – 3 bracket explicitly through every downstream calculation.

2.3 Freeze the two common choices, in writing, before you look at any results. From the inventory, define:

- **The common uv range** `'[uv_lo, uv_hi]'` in $k\lambda$: `'uv_lo'` set by the longest of the short-baseline cut-ons across the sample, and `'uv_hi'` set by the shortest of the well-sampled outer baselines — where "well-sampled" is defined by a fixed criterion you write down now, e.g. the uv distance beyond which a target has fewer than N visibilities per bin or a per-bin expected noise above some threshold. This is the range over which the headline compact fraction is measured for all targets.
- **The common restoring beam** for Observable 2: a circular beam with FWHM equal to the largest delivered major axis in the sample, rounded up to a clean value.

Commit these to a version-controlled config file with a git tag before running the sample. The point of the exercise is a *uniform* census, and uniformity that is chosen after seeing the answers is not uniformity.

0.3 3. Weeks 4–8: building and validating the visibility-binning pipeline

Develop on the two highest-S/N targets from the inventory; do not touch the rest of the sample until injection–recovery passes.

3.1 Continuum MS preparation. For each target, 'split' out the science-target field and the science spws. Channel-average with `'mstransform'` to a small number of channels per spw rather than a single channel — you need enough channels that the frequency-dependent uv coordinate is preserved and bandwidth smearing at the longest baselines is negligible; verify by computing the smearing fraction at `'uv_hi'` and choosing the channel width so it stays below $\sim 1\%$. Time-average similarly (typically 10–30 s), with the time-smearing fraction at `'uv_hi'` checked the same way. Concatenate execution blocks only after checking that their binned amplitude profiles agree within the flux-scale uncertainty; if an EB disagrees, report it and exclude it rather than averaging the discrepancy away.

3.2 Weights. Everything downstream depends on the visibility weights being right, so audit them: for each target compute the empirical scatter of the visibilities in short time/baseline windows and compare with `'1/ $\sqrt{w}$ '`. If the pipeline weights are internally consistent, use them as delivered; if not,

run ‘statwt’ with an identical configuration for every target and document which targets required it. Record the ratio (empirical scatter)/(nominal) per target — it is a term in your error budget.

3.3 Stokes I and phase shift. ALMA has linear feeds, so form ‘ $I = (XX + YY)/2$ ’ with ‘ $w_I = 4/(1/w_{XX} + 1/w_{YY})$ ’. Then find the emission centroid and move it to the phase centre: make a light, shallow ‘tclean’ image, fit the emission with ‘imfit’ (for a compact source) or take the flux-weighted centroid inside a hand-checked mask (for an arc), then apply CASA’s ‘phaseshift’ to put that position at the phase centre. This step is what makes the next one legitimate.

3.4 Binning — real part, not amplitude. Vector-averaged amplitudes are positively biased at low S/N, so you never average ‘ $|V|$ ’. After the phase shift, bin the visibilities in uv distance and take the weighted mean of the **real** part in each bin:

‘ $\text{Re } V_k = \sum w_i \text{Re}(V_i) / \sum w_i$ ’, with ‘ $\sigma_k = (\sum w_i)^{-1/2}$ ’ scaled by the weight-audit factor from 3.2, and cross-checked against the weighted scatter within the bin (‘ $\sigma_{k,\text{emp}}$ ’). Quote the larger of the two. Take the bin abscissa as the weighted mean uv distance within the bin, not the bin centre.

Bin the **imaginary** part identically and report it as a null/asymmetry check: quote χ^2/dof of $\text{Im } V$ against zero per target. A significant imaginary signal means either a residual centroid error or a genuinely asymmetric brightness distribution about the chosen centroid, and you say which after re-running with a perturbed centroid.

Choose bin widths so that each bin reaches comparable expected S/N (approximately equal ‘ $\sum w$ ’), which for typical ALMA coverage means narrow bins at short uv distances and wider ones at long.

3.5 The three numbers per target.

- **Total flux density S_{tot} :** weighted mean of ‘ $\text{Re } V$ ’ over the short-baseline plateau, with the plateau defined by a fixed uv-distance ceiling set in Section 2.3. Cross-check against an aperture-integrated flux from the image; agreement within the flux-scale uncertainty is a required validation, not an optional one.
- **Compact fraction f_c :** ‘ $\text{Re } V$ ’ averaged over the outer bins divided by ‘ S_{tot} ’. Report the value measured over the frozen common range ‘ $[uv_{\text{lo}}, uv_{\text{hi}}]$ ’ as the headline number for all targets, and alongside it the value measured over each target’s own native outer coverage. Both go in the table; the common-range value is what makes the census comparable, the native value is what shows what each dataset could actually see.
- **Characteristic angular scale $\theta_{1/2}$:** the uv distance at which the binned profile falls to half the short-baseline plateau, obtained by linear or monotone-spline interpolation across the binned points, with uncertainty from bootstrap resampling of the visibilities within bins. Label this a *characteristic angular scale*, not a source size, everywhere it appears — in the table header, in the figure axis labels, and in the text. A curved, non-circular source does not invalidate azimuthal binning (the binned profile is a well-defined angle-averaged measure of power on each scale), but it does mean $\theta_{1/2}$ cannot be read as a physical extent, and the injection tests in 3.6 are how you quantify that.

3.6 Injection–recovery through the real uv sampling. Build a deliberately non-Gaussian input: a curved arc — a segment of an annulus with finite width, with and without embedded clumps carrying a specified fraction of the flux. Fourier-transform this model onto the *actual* uv coordinates of a real MS (CASA ‘ft’ with the model image, then copy MODEL_DATA into DATA), and add noise drawn per-visibility from the audited weights. Run the injected dataset through the

entire pipeline of 3.3–3.5 with no manual intervention. Repeat over a grid of: arc curvature and width, clump flux fraction, and total flux scaled to span the S/N range of the real sample. From the recovered-minus-input distributions of ‘f_c’ and ‘ $\theta_{1/2}$ ’, derive (a) the bias as a function of S/N and (b) the scatter floor. That scatter floor is the systematic uncertainty you quote in the table; the statistical error from the weights alone is not the whole story and you should say so. Also perturb the injected centroid by ± 0.5 beam and fold the resulting change in ‘f_c’ and ‘ $\theta_{1/2}$ ’ into the same systematic term.

0.4 4. Weeks 9–16: running the full public sample

Run the frozen pipeline over every public target with no per-target tuning. One directory per MOUS, one config file, all CASA logs retained, all table entries emitted by script. Where a target is a non-detection or is dominated by structure smoothed out beyond ‘uv_hi’, that is a result: report ‘f_c’ with its uncertainty (including values consistent with zero) and state the constraint as “the emission is dominated by structure resolved out beyond the common uv cut.” Do not drop targets for having uninteresting answers; the census is the deliverable and a quantitative upper limit on compactness is a measurement.

0.5 5. Weeks 17–19: brightness-temperature maps and the peak-T_B table

5.1 Imaging, twice. Produce two image sets per target with ‘tclean’, identical parameters across the sample: ‘specmode=’mfs’, ‘deconvolver=’mtmfs’ with ‘nterms’ chosen from the fractional bandwidth, a fixed Briggs robust, cell size $\approx$ beam/5, ‘auto-multithresh’ masking with fixed parameters, cleaned to $\sim 2\sigma$, and primary-beam corrected.

- *Native set:* at each target’s delivered resolution.
- *Common-beam set:* apply a per-target ‘uvtaper’ chosen to bring the fitted beam to the frozen common circular beam and set ‘restoringbeam’ to that beam explicitly, then verify from the image header that the beam is what you asked for. Do not simply ‘imsmooth’ the native images — that matches the restoring beam without matching the uv weighting.

5.2 Convert peak intensity to brightness temperature with the full Planck function.

Take the peak pixel value ‘S_peak’ (Jy/beam) from the pbcor image, convert to specific intensity with ‘ $I_\nu = S_{\text{peak}} / \Omega_{\text{beam}}$ ’, ‘ $\Omega_{\text{beam}} = \pi \theta_{\text{maj}} \theta_{\text{min}} / (4 \ln 2)$ ’. Then invert the *full* Planck function — not Rayleigh–Jeans, which fails at these rest frequencies:

$$T_B(\nu_{\text{obs}}) = (h\nu_{\text{obs}}/k) / \ln(1 + 2h\nu_{\text{obs}}^3 / (c^2 I_\nu))$$

with ‘ ν_{obs} ’ the measured band-centre frequency from the inventory (Section 2.2). Because ‘ I_ν/ν^3 ’ is a lensing invariant, **no magnification correction is applied** — state this explicitly as the reason. Convert to the rest frame with ‘ $T_{B,\text{rest}} = (1+z) T_B(\nu_{\text{obs}})$ ’, evaluated at ‘ $\nu_{\text{rest}} = (1+z) \nu_{\text{obs}}$ ’; where a per-source redshift exists in the project metadata, use it, and where it does not, tabulate the pair of values at $z = 1$ and $z = 3$ as an explicit bracket rather than assuming a fiducial redshift.

5.3 Report every value as a lower bound. The peak ‘T_B’ you measure is degraded both by beam filling factor (the beam is almost certainly larger than the emitting structure) and by optical depth below unity, so each entry is a lower bound on the true peak, and the paper says so at every quotation. Alongside each value, quote the CMB temperature at that redshift, ‘ $T_{\text{CMB}}(z) = 2.725 (1+z)$ ’ K, as the background/contrast floor — the interferometer measures contrast against

the sky, so the CMB sets the level against which your ‘T_B’ is a contrast, and readers need that number in the same row to interpret a low value. A result of ”peak rest-frame T_B is only a few K” is a genuine joint bound on filling factor and optical depth; present it that way rather than as a null.

5.4 Tabulate at both resolutions. Peak ‘T_B,rest’ at native beam and at the common beam, both in the table. The common-beam column is the one that is comparable across targets with different delivered beams; the native column shows what each dataset actually resolved.

0.6 6. Error budget

Carry four terms explicitly, per target: (i) statistical, from the audited visibility weights; (ii) the injection-calibrated systematic floor on ‘f_c’ and ‘ $\theta_{1/2}$ ’ from Section 3.6, including the centroid perturbation term; (iii) the absolute flux-scale accuracy quoted in each MOUS’s QA2 report for its band, which affects ‘S_tot’ and ‘T_B’ but cancels in the ratio ‘f_c’; (iv) for ‘T_B,rest’, the redshift term, shown as the $z = 1$ to $z = 3$ bracket where no per-source redshift exists. State which terms cancel where — ‘f_c’ being a flux ratio is one of the reasons it is the robust headline number.

0.7 7. Deliverables, bookkeeping, and schedule

The primary deliverable is one table with one row per public target: total flux density; compact fraction at the common uv cut and at native coverage; characteristic angular scale with injection-calibrated uncertainty; peak rest-frame ‘T_B’ at native and common beam with the redshift bracket shown; plus the Im V null-check χ^2/dof . The secondary deliverable is the figure set of binned uv-amplitude profiles, one panel per target, with the common uv range shaded on every panel.

Everything is scripted and version-controlled: a git repository with the frozen config tagged at week 3, per-MOUS working directories, retained CASA logs, and a build script that regenerates the table and all figures from the measurement products. **Freeze all measurements at week 20** and spend weeks 20–24 on the write-up; the write-up is a scheduled block, not whatever time is left over. The schedule is: weeks 1–3 archive availability and inventory and the written parameter freeze; weeks 4–8 pipeline build and injection–recovery on the two best targets; weeks 9–16 the full public sample; weeks 17–19 brightness temperatures; weeks 20–24 writing.

If and only if the census is frozen on schedule at week 20, add a qualitative morphological comparison of one target’s continuum against the archival rest-frame optical HST imaging that the proposal itself names for the dust-extinction-geometry question — overlay contours, describe what you see, make no quantitative claim. This is a contingency block, not required scope, and it is dropped without discussion if week 20 slips.

The paper must carry the required acknowledgement verbatim: the ADS/JAO.ALMA#2022.1.01311.S ALMA partnership statement, plus the standard NRAO acknowledgement for North American authors.